

David Fernández Alejo

Platform & Security Engineer

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Summary

Platform and Security Engineer with 5 years at GMV in the space sector (Galileo programme), operating services in mission-critical, controlled, and air-gapped environments. Focused on Apache Kafka platforms (KRaft, TLS/SASL, Schema Registry, MirrorMaker 2), identity and access management (LDAP, Kerberos, OAuth2, PKI), system hardening, and automation. SAFe Team Coach since 2024. M.Sc. in Cybersecurity.

Skills

Streaming & Messaging: Apache Kafka (KRaft), TLS/SASL (PLAIN, OAUTHBEARER), MirrorMaker 2, Confluent Schema Registry (Avro, JSON)

Security & IAM: PKI, TLS, LDAP, Kerberos, OAuth2, system hardening, security baselines

Infrastructure: Linux, Windows Server, VMware ESXi, HPE iLO, Docker, Docker Compose, Kubernetes (fundamentals), Vagrant

Automation & CI/CD: Ansible, Bash, Jenkins, Rundeck, Git

Monitoring & SIEM: Splunk, Grafana, Wireshark

Offensive Security & Forensics: Metasploit, IDA, Jadx, apktool, Autopsy, CAINE, RegRipper, Hayabusa

Programming: Go, C++, C, Python, Java, JavaScript (React), SQL

Standards & Methods: ECSS, SAFe

Experience

GMV – Galileo Project

Software Engineer

SAFe Team Coach (alongside engineering role)

Sep 2021 – Present

2024 – Present

Apache Kafka platform

- Designed and implemented end-to-end Apache Kafka architectures: KRaft cluster configuration, topic management, acknowledgments, quotas, and performance tuning.
- Secured Kafka with TLS encryption and SASL authentication (PLAIN, OAUTHBEARER).
- Configured and managed Confluent Schema Registry (Avro, JSON) in both connected and air-gapped environments.
- Developed Kafka producers and consumers in Go with Confluent libraries, including custom message formats and partitioning strategies.
- Set up cross-cluster replication with MirrorMaker 2.
- Troubleshoot and optimized broker configuration, connectivity, and client behavior in distributed environments.

Secure infrastructure and automation

- Configured and secured core infrastructure services (DNS, LDAP, PKI, firewalling, networking) in controlled and security-sensitive environments.
- Applied security hardening guidelines across multiple systems, aligning with security baselines and compliance requirements.
- Operated services on VMware ESXi virtualized infrastructure, including out-of-band server management via HPE iLO.
- Contributed to Jenkins CI/CD pipelines for automated deployments.
- Troubleshoot complex system-level issues across networking, authentication, and service integration layers.

Mission-critical software

- Integrated and validated software modules in embedded systems for satellite navigation, ensuring real-time performance and deterministic behavior.
- Worked to ECSS space engineering standards, ensuring high levels of reliability, safety, and security.

Team coaching (SAFe)

- Acting as SAFe Team Coach, driving agile adoption, facilitating ceremonies, and improving delivery in cross-functional teams.

Personal Projects

Self-directed work, under continuous development. Focused on self-hosting, secure network exposure, and backend design; repositories kept private.

Self-hosted infrastructure

Ubuntu Server, Docker Compose, Tailscale, Cloudflare Tunnel

- Running containerized services on a personal Linux server to practice deployment, service isolation, and least-privilege exposure: private access over a mesh VPN, public access through a tunnel instead of open inbound ports, and a proxy mediating container access to the Docker socket.

Backend and API design

Python, FastAPI, SQLAlchemy, SQLite, React PWA

- Building an API-first application to explore domain modeling, schema migrations, and deterministic calculation engines designed as pure functions, so results can be fully recomputed from source data rather than stored as state.

Data collection and observability

Python, TimescaleDB, Grafana, Docker Compose

- Automated collection pipeline feeding a time-series database, with scheduled jobs and Grafana dashboards. An interest in instrumenting systems, keeping historical data, and making trends visible.

Education

M.Sc. in Cybersecurity – IMF Smart Education (with Deloitte)

2023 – 2025

- Master's thesis: CTF development. Report: github.com/david10923/Master/TFM
- Hands-on projects in cyber intelligence, ethical hacking (Metasploit), secure development, SIEM (Splunk), reverse engineering (IDA, Jadx, apktool), mobile security, and digital forensics (Autopsy, CAINE, RegRipper, Hayabusa). Reports: github.com/david10923/Master

B.Sc. in Software Engineering – Universidad Complutense de Madrid

2017 – 2022

- Java-focused curriculum: object-oriented design, design patterns, and application development. Projects: github.com/david10923

Additional training: Kubernetes courses (PUE, Udemy)

Languages

Spanish (native) | English (C1) | French (B2)